increase the accuracy of the model in associating the correct class or value to new entries. Examples of SL are object detection (classification) or predictive models to estimate goods' prices (regression).

- **Reinforcement learning**: rather than providing explicit labels associated to the data, the model performs some kind of task and, only at the end, receives feedback about the outcome (stating either success or failure). This is one of the most interesting area of research at the time of writing, and some examples include game theory (for instance, an agent learning to play chess, or Go) and many areas of robotics (like teaching a robotic arm to joggle).
- **Unsupervised learning**: this category differs from the first two because in this case the algorithms are not presented with any feedback on data, but their goal is rather making sense of data by extrapolating its inner (and often hidden) structure. Clustering is the main form of unsupervised learning.

We will obviously focus on unsupervised learning in the rest of the chapter: clustering algorithms, in fact, take an unlabeled dataset, and try to gather as much information as possible about its structure but grouping together similar data points, while setting apart dissimilar ones.

Although at first it might seem less intuitive than supervised or reinforcement learning, **clustering** has several natural applications, and the possibly the best way to describe clustering is by exemplifying a few of them:

1. **Market segmentation**: take purchases data, and find out group of similar customers. Since they behave in